ETF Forecaster - Methodology and Results

Generated 2026-09-02 12:35 - mo-etf-forecaster - universe of 40 tickers, horizons 1-5 days

1. Methodology (design of record)

Objective. For each of 40 ETFs/indices, forecast the probability that the close in h trading days (h = 1..5) is above today's close, plus a conformal quantile distribution of the h-day log return, and maintain an audited out-of-sample track record.

Data. Full-history daily bars (yfinance) for the universe plus the CBOE volatility complex (VIX, VIX9D, VIX3M, VIX6M, VVIX, SKEW, VXN, VXD, OVX, GVZ) and 10 cross-asset series; 14 fast FRED series (yields, curve, credit OAS, dollar, oil, NFCI, Fed balance sheet, SOFR); 24 slow macro series pulled as point-in-time ALFRED vintages (CPI, PCE, INDPRO, capacity utilization, payrolls, claims, M2, GDP, housing, sentiment, NBER dates); Shiller S&P 500 earnings/CAPE; per-ticker 30-day implied and historical volatility from IBKR; and a daily option-chain snapshot out to 120 DTE from which 25-delta risk reversal, butterfly, ATM term structure and a dealer-gamma proxy are computed (this block accrues history forward from first snapshot).

Point-in-time discipline. Slow macro features are computed from the series exactly as known on each historical date via ALFRED vintages (fallback: conservative publication lags). Weekly FRED series are lagged one week. Korean closes are same-day usable for US targets but US closes are lagged one day for Korean targets. Weekly/monthly chart patterns only become visible the first daily bar after their (completed) higher-timeframe bar ends. Shiller earnings are lagged three months.

Features. ~690 columns per ticker in 12 blocks: PRICE (always on; returns, gaps, range-vol estimators, last-60-candle block), TECH (20+ indicators), PATTERN_D1 and PATTERN_HTF (16 chart-pattern detectors - double/triple tops and bottoms, head and shoulders, channels, S/R, wedges, flags, trendlines and breaks - run on daily, weekly and monthly bars, converted to per-bar features with direction decay, trigger flags, ATR distances and cross-timeframe agreement), REGIME (ADX state, HH/HL market structure, variance ratio, MA posture, vol terciles, leak-free Markov-switching probability), VOLOPT (VIX complex, term slopes, IBKR IV, VRP, HAR-RV, EWMA/GARCH), SKEW (chain-derived), MACRO_FAST, MACRO_SLOW (vintage-aligned), EARNINGS, CROSS (session-aligned lead-lag, relative strength, beta, group momentum rank), SEASON (calendar and event days).

Search. A staged include/exclude ablation over the 11 toggleable blocks per (ticker-group, horizon): stage 1 screens each block against a PRICE-only floor; stage 2 greedy forward/backward selection; stage 3 exhaustive 2^k over the shortlist; stage 4 model/hyperparameter search (logistic, elastic-net, LightGBM, XGBoost, CatBoost, RF/ET/HGB, LSTM/TCN over the candle tensor; Optuna TPE in full mode) including the lag-window X in {10, 20, 40, 60}; stage 5 stacking of the top configurations with isotonic calibration. All selection is nested in the development slice (first 60% of history). Every evaluated cell is persisted with its running trial count.

Validation. Purged, embargoed, expanding walk-forward: training data for a refit forecasting date t ends h bars before t (both purge and embargo for overlapping h-day labels). The out-of-sample track record consists exclusively of predictions dated after the development slice; champion switching along the way uses only trailing OOS data available at that moment (sticky rule). Magnitude quantiles are conformalized (CQR) on a held-out calibration slice targeting 80% coverage for the q10-q90 band. Reported results carry Benjamini-Hochberg FDR flags and binomial p-values; hit rates should be read net of the recorded trial counts.

Honest expectations. Daily equity-index direction is a low-signal problem: a realistic sustained OOS hit rate is 52-56%, not 70%. VIX-family tickers score much higher hit rates for structural reasons (persistent downward drift of VIXY, mean reversion of VIX), which the probability calibration reflects. The chain-derived SKEW block has essentially no history yet and will only earn a place in the ablation as snapshots accrue.

2. Data dictionary and coverage

Daily bars (yfinance), full available history:

ticker	rows	start	end
SPY	8456	1993-01-29	2026-09-02
QQQ	6914	1999-03-10	2026-09-02
IWM	6606	2000-05-26	2026-09-02
MDY	7884	1995-05-04	2026-09-02
DIA	7199	1998-01-20	2026-09-02
XLK	6965	1998-12-22	2026-09-02
XLF	6966	1998-12-22	2026-09-02
XLE	6966	1998-12-22	2026-09-02
XLB	6965	1998-12-22	2026-09-02
XLI	6965	1998-12-22	2026-09-02
XLV	6966	1998-12-22	2026-09-02
XLY	6965	1998-12-22	2026-09-02
XLP	6965	1998-12-22	2026-09-02
XLC	2062	2018-06-19	2026-09-02
XLU	6965	1998-12-22	2026-09-02
XLRE	2741	2015-10-08	2026-09-02
^VIX	9235	1990-01-02	2026-09-02
^VVIX	4938	2007-01-03	2026-09-02
VIXY	3939	2011-01-04	2026-09-02
SOXX	6322	2001-07-13	2026-09-02
SMH	6601	2000-06-05	2026-09-02
MU	10646	1984-06-01	2026-09-02
000660.KS	6665	2000-01-04	2026-09-01
005930.KS	6665	2000-01-04	2026-09-01
EWG	7665	1996-03-18	2026-09-02
EWQ	7664	1996-03-18	2026-09-02
EWJ	7664	1996-03-18	2026-09-02
EWU	7665	1996-03-18	2026-09-02
EWC	7665	1996-03-18	2026-09-02
EEM	5884	2003-04-14	2026-09-02
FXI	5510	2004-10-08	2026-09-02
EWY	6615	2000-05-12	2026-09-02
EWT	6587	2000-06-23	2026-09-02
INDA	3666	2012-02-03	2026-09-02

ticker	rows	start	end
EWZ	6572	2000-07-14	2026-09-02
EWW	7665	1996-03-18	2026-09-02
^GSPTSE	11846	1979-06-29	2026-09-02
XFN.TO	6385	2001-03-29	2026-09-02
XEG.TO	6390	2001-03-23	2026-09-02
XMA.TO	5191	2005-12-28	2026-09-02
^VIX	9235	1990-01-02	2026-09-02
^VIX9D	3908	2011-01-03	2026-09-02
^VIX3M	5033	2006-07-17	2026-09-02
^VIX6M	4665	2008-01-02	2026-09-02
^VVIX	4938	2007-01-03	2026-09-02
^SKEW	9160	1990-01-02	2026-09-01
^VXN	6440	2001-01-23	2026-09-02
^RVX	0	nan	nan
^VXD	5189	2005-11-22	2026-09-02
^OVX	4859	2007-05-10	2026-09-02
^GVZ	4591	2008-06-03	2026-09-02
^TNX	16154	1962-01-02	2026-09-02
^TYX	12413	1977-02-15	2026-09-02
DX-Y.NYB	14135	1971-01-04	2026-09-02
CL=F	6535	2000-08-23	2026-09-02
GC=F	6526	2000-08-30	2026-09-02
HG=F	6531	2000-08-30	2026-09-02
TLT	6062	2002-07-30	2026-09-02
HYG	4881	2007-04-11	2026-09-02
LQD	6062	2002-07-30	2026-09-02
RSP	5872	2003-05-01	2026-09-02

Other artifacts (vol indices, macro, vintages, IBKR IV, skew):

artifact	rows	start	end
calendar/events	8456	1993-01-29	2026-09-02
cross/CL_F	6535	2000-08-23	2026-09-02
cross/DX-Y_NYB	14135	1971-01-04	2026-09-02
cross/GC_F	6526	2000-08-30	2026-09-02
cross/HG_F	6531	2000-08-30	2026-09-02
cross/HYG	4881	2007-04-11	2026-09-02
cross/LQD	6062	2002-07-30	2026-09-02

artifact	rows	start	end
cross/RSP	5872	2003-05-01	2026-09-02
cross/TLT	6062	2002-07-30	2026-09-02
cross/TNX	16154	1962-01-02	2026-09-02
cross/TYX	12413	1977-02-15	2026-09-02
fred/fast	9566	1990-01-02	2026-09-01
ibkr/DIA_hv	3757	2011-09-07	2026-09-01
ibkr/DIA_iv	3768	2011-09-07	2026-09-02
ibkr/EEM_hv	3757	2011-09-07	2026-09-01
ibkr/EEM_iv	3768	2011-09-07	2026-09-02
ibkr/EWC_hv	3757	2011-09-07	2026-09-01
ibkr/EWC_iv	3768	2011-09-07	2026-09-02
ibkr/EWG_hv	3757	2011-09-07	2026-09-01
ibkr/EWG_iv	3767	2011-09-07	2026-09-02
ibkr/EWJ_hv	3758	2011-09-06	2026-09-01
ibkr/EWJ_iv	3768	2011-09-06	2026-09-02
ibkr/EWQ_hv	3757	2011-09-07	2026-09-01
ibkr/EWQ_iv	3756	2011-09-07	2026-09-02
ibkr/EWT_hv	3758	2011-09-06	2026-09-01
ibkr/EWT_iv	3768	2011-09-06	2026-09-02
ibkr/EWU_hv	3757	2011-09-06	2026-09-01
ibkr/EWU_iv	3689	2011-09-07	2026-09-02
ibkr/EWW_hv	3757	2011-09-07	2026-09-01
ibkr/EWW_iv	3768	2011-09-07	2026-09-02
ibkr/EWY_hv	3757	2011-09-07	2026-09-01
ibkr/EWY_iv	3767	2011-09-07	2026-09-02
ibkr/EWZ_hv	3757	2011-09-07	2026-09-01
ibkr/EWZ_iv	3767	2011-09-07	2026-09-02
ibkr/FXI_hv	3757	2011-09-07	2026-09-01
ibkr/FXI_iv	3767	2011-09-07	2026-09-02
ibkr/INDA_hv	3636	2012-03-19	2026-09-01
ibkr/INDA_iv	3247	2013-07-18	2026-09-02
ibkr/IWM_hv	3757	2011-09-07	2026-09-01
ibkr/IWM_iv	3768	2011-09-07	2026-09-02
ibkr/MDY_hv	3757	2011-09-07	2026-09-01
ibkr/MDY_iv	3757	2011-09-07	2026-09-02
ibkr/MU_hv	3757	2011-09-07	2026-09-01

artifact	rows	start	end
ibkr/MU_iv	3768	2011-09-07	2026-09-02
ibkr/QQQ_hv	3758	2011-09-06	2026-09-01
ibkr/QQQ_iv	3769	2011-09-06	2026-09-02
ibkr/SMH_hv	3652	2012-02-06	2026-09-01
ibkr/SMH_iv	3691	2011-12-22	2026-09-02
ibkr/SOXX_hv	3757	2011-09-07	2026-09-01
ibkr/SOXX_iv	3690	2011-09-07	2026-09-02
ibkr/SPY_hv	3757	2011-09-07	2026-09-01
ibkr/SPY_iv	3767	2011-09-07	2026-09-02
ibkr/VIXY_hv	3750	2011-09-06	2026-09-01
ibkr/VIXY_iv	3763	2011-09-06	2026-09-02
ibkr/XLB_hv	3757	2011-09-07	2026-09-01
ibkr/XLB_iv	3768	2011-09-07	2026-09-02
ibkr/XLC_hv	2025	2018-08-01	2026-09-01
ibkr/XLC_iv	2060	2018-06-22	2026-09-02
ibkr/XLE_hv	3757	2011-09-07	2026-09-01
ibkr/XLE_iv	3768	2011-09-07	2026-09-02
ibkr/XLF_hv	3757	2011-09-07	2026-09-01
ibkr/XLF_iv	3766	2011-09-07	2026-09-02
ibkr/XLI_hv	3757	2011-09-07	2026-09-01
ibkr/XLI_iv	3768	2011-09-07	2026-09-02
ibkr/XLK_hv	3757	2011-09-07	2026-09-01
ibkr/XLK_iv	3768	2011-09-07	2026-09-02
ibkr/XLP_hv	3757	2011-09-07	2026-09-01
ibkr/XLP_iv	3768	2011-09-07	2026-09-02
ibkr/XLRE_hv	2703	2015-11-19	2026-09-01
ibkr/XLRE_iv	2719	2015-10-27	2026-09-02
ibkr/XLU_hv	3757	2011-09-07	2026-09-01
ibkr/XLU_iv	3768	2011-09-07	2026-09-02
ibkr/XLV_hv	3757	2011-09-07	2026-09-01
ibkr/XLV_iv	3768	2011-09-07	2026-09-02
ibkr/XLY_hv	3757	2011-09-07	2026-09-01
ibkr/XLY_iv	3768	2011-09-07	2026-09-02
shiller/shiller	1845	1871-01-31	2024-09-30
skew/DIA	1	2026-09-02	2026-09-02
skew/EEM	1	2026-09-02	2026-09-02

artifact	rows	start	end
skew/EWC	1	2026-09-02	2026-09-02

Fast FRED series: DGS3MO, DGS2, DGS10, T10Y2Y, T10Y3M, DFII10, T10YIE, BAMLH0A0HYM2, BAMLC0A0CM, DTWEXBGS, DCOILWTICO, NFCI, WALCL, SOFR.
ALFRED vintage series: CPIAUCSL, CPILFESL, PCEPI, PCEPILFE, MICH, INDPRO, IPMAN, TCU, NEWORDER, HOUST, PERMIT, UNRATE, U6RATE, PAYEMS, ICSA, CCSA, SAHMREALTIME, M2SL, WM2NS, M2REAL, GDPC1, UMCSENT, USREC, CP.

3. Feature blocks

block	prefix	toggleable	columns (SPY)	description
PRICE	px_	False	112	returns, gaps, range-vol estimators, last-X candles
TECH	tech_	True	39	top-20 technical indicators
PATTERN_D1	patd_	True	53	chart patterns on daily bars
PATTERN_HTF	path_	True	105	chart patterns on weekly/monthly bars + agreement
REGIME	reg_	True	15	trend/range state, market structure, Markov regime
VOLOPT	vol_	True	35	VIX complex, term slopes, IBKR IV, HAR/GARCH
SKEW	skw_	True	15	chain-derived rr25/bf25/term/GEX (short history)
MACRO_FAST	mf_	True	58	daily FRED: yields, curve, credit, dollar, oil
MACRO_SLOW	ms_	True	132	vintage-aligned CPI, INDPRO, UNRATE, claims, M2, ...
EARNINGS	earn_	True	9	Shiller earnings growth, CAPE, Fed-model spread
CROSS	x_	True	102	session-aligned cross-asset lead-lag and ranks
SEASON	cal_	True	19	day-of-week, turn-of-month, OPEX, FOMC, releases

4. Grid-search design and budgets

Horizons: [1, 2, 3, 4, 5]. Walk-forward: min_train 1000 bars, refit every 21 (full) / 63 (fast) bars, embargo = horizon, calibration slice 15%. Ablation candidates: TECH, PATTERN_D1, PATTERN_HTF, REGIME, VOLOPT, SKEW, MACRO_FAST, MACRO_SLOW, EARNINGS, CROSS, SEASON. Greedy min gain 0.0005, exhaustive cap 2^6. Optuna trials per cell (full): 40. Window grid: [10, 20, 40, 60]. Champion metric log_loss with sticky margin 0.002; BH-FDR at 10%.

5. Ablation results - which blocks earn their keep

No ablation results yet - run train_search.

6. Out-of-sample performance

No OOS history yet.

7. Known limitations

(1) The chain-derived SKEW block starts with a single day of history; its value is unmeasurable for months. (2) IBKR implied vol covers ~15 years for large ETFs and less for younger ones; earlier history has no per-ticker IV, only the index complex. (3) ALFRED vintages typically begin 1996-1998; before that, publication-lag shifts are used. (4) The fast-mode backfill uses reduced budgets (annual refits in the nested search, quarterly in the outer walk-forward); the scheduled monthly full ablation re-runs everything at full budget and results supersede. (5) Sequence models (LSTM/TCN) participate only in the full-budget roster. (6) A ~52-55% pooled hit rate on daily horizons is expected; treat any ticker cell that has not survived BH-FDR as noise. (7) ^GSPTSE has no options and the .KS/.TO names have thinner coverage; their SKEW/VOLOPT blocks degrade to the index-level complex.